BT 5240 Computational Systems Biology

Constraint-based Modelling for the study of Polymicrobial Infections

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Project Abstract

Polymicrobial Diseases result from concurrent infections caused by multiple pathogen species. The infection caused by one pathogen may create a favourable niche for another pathogen, or two pathogens infecting a host independently can lead to a particular disease [1]. For example, *Vibrio cholerae* and Enterotoxic *Escherichia coli* (ETEC) are known to coinfect the small intestine epithelial cells in humans.

Studying the metabolism of coinfecting pathogens is important to understand how they differ from their non-infectious states, since the metabolism of pathogens is known to be important for their virulence and survival in the host environment.

Genome-scale metabolic models (GSMMs) have been used to study the integrated host-pathogen metabolic networks [2]. Moreover, GSMMs have been employed to study the metabolism of pathogens during co-infections in culture media [3]. However, models that integrate the host metabolic network with the metabolic networks of multiple coinfecting pathogens in polymicrobial diseases have not been investigated.

Therefore, I propose to utilize previously published GSMMs of the human small intestine epithelial cells [4], *Vibrio cholerae*, and ETEC [3] to construct an integrated metabolic network comprising the host cell and coinfecting pathogens. Such an integration of existing GSMMs entails the following three steps:

1. Setting up the integrated Stoichiometric Matrix using individual GSMMs.
2. Renaming reactions and metabolites from each organism, and introducing new exchange reactions, if necessary.
3. Defining the objective function for the new Flux Balance Analysis (FBA) problem:

   Due to the lack of data from *in vivo* models of coinfections of human small intestine epithelial cells, I propose to explore various objective functions (such as maximising the pathogen biomass reactions, utilising parsimonious FBA to optimise reaction fluxes in the pathogens' metabolic networks, etc., as proposed in [5]) and investigate which metabolites and reactions are important for the growth of the pathogens within the host environment.

The results from performing FBA using this model will be compared with those obtained from the GSMM of a healthy host cell and coinfection models of the pathogens to study how the metabolic capabilities of the host and the pathogens differ across different states.

Furthermore, the model will be used to predict essential genes for the coinfecting pathogens and compare these predictions with known essential genes for individual pathogens [3].

Thus, this project aims to use the integrated host-coinfection model to study the metabolic interactions between the host and the coinfecting pathogens in polymicrobial diseases.

References:

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